

Bootstrap — Topics 6-14: Utility Classes

Study paper format: notes (Roman Urdu + English technical terms) + active recall Q&A at the end. Covers: Spacing, Sizing, Borders, Shadows, Display, Flex, Position, Overflow, Visibility.

6. Spacing ★ ★ ★

Spacing utilities margin aur padding **bina custom CSS** ke control karne dete hain.

Naming Pattern

```
{property}{sides}-{size}
```

- **property:** **m** (margin) ya **p** (padding)
- **sides:** **t** (top), **b** (bottom), **s** (start/left), **e** (end/right), **x** (left+right), **y** (top+bottom), blank (all sides)
- **size:** **0** se **5** (aur **auto** margin ke liye)

Margin

```
m-0      → margin: 0 (koi margin nahi)
mt-5     → margin-top: 5px
mx-auto   → left-right margin auto (horizontal centering ke liye)
mb-3     → margin-bottom: 3px
```

html

```
<div class="mt-5">Top margin bara hai</div>
<div class="mx-auto" style="width: 200px;">Horizontally centered</div>
```

mx-auto element ko **horizontally center** karta hai (element ki width fixed honi chahiye).

Padding

```
p-3    → sab sides par medium padding  
py-5   → top + bottom par bara padding  
px-2   → left + right par choti padding
```

html

```
<div class="p-3 bg-light">Padding on all sides</div>  
<div class="py-5">Top-bottom bara padding</div>
```

Responsive Spacing

Screen size ke hisab se spacing change ki ja sakti hai:

```
mt-lg-5 → sirf large screens (≥992px) par top margin bara ho
```

html

```
<div class="mt-3 mt-lg-5">  
    Mobile par choti margin, desktop par bari margin.  
</div>
```

7. Sizing

Sizing utilities element ki width/height ko **percentage** mein control karte hain (parent ke relative).

Width

```
w-25    → 25% width
w-50    → 50% width
w-75    → 75% width
w-100   → 100% width (full width)
```

html

```
<div class="w-50 bg-primary">Half width</div>
```

Height

```
h-25    → 25% height
h-50    → 50% height
h-100   → 100% height
```

Height percentages parent ki height defined honi chahiye tabhi properly kaam karti hain.

Viewport

```
vh-100    → full viewport height (100vh)
min-vh-100 → minimum height = full viewport (content zyada ho to barh sakta hai)
```

html

```
<div class="vh-100 bg-dark text-white">
  Full screen height section
</div>
```

Use case: Hero sections jo pura screen cover karein.

8. Borders

Border utilities elements par border add/remove aur rounded corners dene ke liye hoti hain.

<code>border</code>	→ default border sab sides par
<code>border-0</code>	→ border remove
<code>border-primary</code>	→ border color blue (theme color)
<code>rounded</code>	→ thora rounded corners
<code>rounded-circle</code>	→ poora circle (element ka width = height hona chahiye)
<code>rounded-pill</code>	→ pill shape (buttons/badges ke liye common)

html

```
<div class="border border-primary p-3">Bordered box</div>

<button class="btn btn-primary rounded-pill">Pill Button</button>
```

Note: `border-{color}` sirf border ka color set karta hai (jaise `border-danger`, `border-success`) — theme colors ke sath combine hoti hai.

9. Shadows

Shadow utilities elements ko depth (box-shadow) dete hain.

<code>shadow</code>	→ medium/default shadow
<code>shadow-sm</code>	→ chota shadow
<code>shadow-lg</code>	→ bara shadow

html

```
<div class="shadow p-3 mb-3 bg-white">Default shadow</div>
<div class="shadow-sm p-3 mb-3 bg-white">Small shadow</div>
<div class="shadow-lg p-3 mb-3 bg-white">Large shadow</div>
```

Use case: Cards, modals, dropdowns ko visually lift karne ke liye.

10. Display Utilities

Display utilities CSS `display` property control karti hain — kaise element render ho.

```
d-block    → display: block (full width, apni line)
d-inline   → display: inline (sirf content jitni width)
d-flex     → display: flex (flexbox container)
d-grid     → display: grid (grid container)
d-none     → display: none (element hide ho jata hai)
```

html

```
<div class="d-flex">
  <div>Item 1</div>
  <div>Item 2</div>
</div>

<p class="d-none">Ye text hidden hai</p>
```

Responsive Display

Screen size ke hisab se element show/hide ho sakta hai:

```
d-md-none  → medium screens aur upar element hide ho jaye
d-lg-block → large screens par element block ban jaye (show ho)
```

html

```
<div class="d-block d-lg-none">  
  Ye sirf mobile/tablet par dikhega, desktop par hide ho jayega  
</div>
```

Common pattern: Mobile menu icon (`d-lg-none`) (sirf mobile par dikhe), desktop nav (`d-none d-lg-flex`) (sirf desktop par dikhe).

11. Flex Utilities ★ ★ ★

Flexbox layouts banane ke liye Bootstrap ka powerful utility set. Pehle parent par (`d-flex`) lagana zaroori hai.

Direction

```
flex-row    → items horizontally arrange (default)  
flex-column → items vertically arrange (stacked)
```

html

```
<div class="d-flex flex-column">  
  <div>Item A</div>  
  <div>Item B</div>  
</div>
```

Justify (main axis alignment)

justify-content-center → items center mein
justify-content-between → items ke beech equal space (pehla/aakhri edge par)
justify-content-around → har item ke around equal space
justify-content-evenly → sab spaces (including edges) bilkul equal

html

```
<div class="d-flex justify-content-between">  
  <div>Left</div>  
  <div>Right</div>  
</div>
```

Align (cross axis alignment)

align-items-center → sab items cross-axis par center
align-self-end → sirf ek specific item ko end par align

html

```
<div class="d-flex align-items-center" style="height: 100px;">  
  <div>Vertically centered</div>  
</div>
```

Wrap

flex-wrap → items agar container mein na samayein to next line par wrap ho jayein

Grow

`flex-grow-1` → item available extra space le le (fill remaining space)

Shrink

`flex-shrink-0` → item shrink na ho, apni original size maintain kare

Gap

`gap-3` → flex items ke beech consistent spacing (margin ki jagah)

html

```
<div class="d-flex gap-3">
  <button class="btn btn-primary">One</button>
  <button class="btn btn-secondary">Two</button>
</div>
```

Note: `gap-*` modern aur cleaner approach hai items ke beech spacing dene ka, individual `me-*`/`mb-*` margins lagane ki bajaye.

12. Position Utilities

CSS `position` property control karne ke liye.

position-relative → element apni normal position ke relative offset ho sakta hai
position-absolute → element parent (jo relative/absolute ho) ke relative position ho
position-fixed → element viewport ke relative fixed rehta hai (scroll karne par bhi)
position-sticky → element scroll tak normal rehta hai, phir stick ho jata hai

Top / Start / Translate

top-0 → element ko top edge par align
start-50 → element ko horizontally 50% se position karna
translate-middle → element ko exactly center karne ke liye (top-50/start-50 ke sath use hota hai)

```
html

<div class="position-relative">
  <span class="position-absolute top-0 start-50 translate-middle badge bg-c
    4
  </span>
</div>
```

Ye pattern notification badges ke liye bohat common hai (icon ke top-right corner par number badge).

13. Overflow

Content jab container se bara ho jaye to uska behavior control karti hain.

overflow-auto → agar zaroorat ho to scrollbar aa jaye
overflow-hidden → extra content chhup jata hai (clip ho jata hai)
overflow-scroll → hamesha scrollbar dikhti hai (chahe content fit ho ya na ho)

html

```
<div class="overflow-auto" style="height: 100px;">  
  Bohat sara lamba content jo scroll ho sakta hai...  
</div>
```

14. Visibility

Element ka visibility toggle karti hain — `d-none` se different (yahan element apni space occupy karta rehta hai).

visible → element dikhta hai (default)
invisible → element invisible hota hai, lekin layout mein apni space still occupy karta hai

html

```
<div class="invisible">Ye dikhega nahi, lekin space occupy karega</div>
```

Key difference: `d-none` element ko completely layout se hata deta hai (no space), jabke `invisible` element ki space reserve rehti hai bas element khud nazar nahi aata.

Cheat Sheet

Section Key Classes

Spacing `m-*`, `p-*`, `mt-*`, `px-*`, `mx-auto`, responsive: `mt-lg-5`

Sizing `w-25/50/75/100`, `h-25/50/100`, `vh-100`, `min-vh-100`

Borders `border`, `border-0`, `border-primary`, `rounded`, `rounded-circle`, `rounded-pill`

Shadows `shadow`, `shadow-sm`, `shadow-lg`

Display `d-block`, `d-inline`, `d-flex`, `d-grid`, `d-none`, responsive: `d-md-none`

Flex `flex-row/column`, `justify-content-*`, `align-items-*`, `align-self-*`, `flex-wrap`, `flex-grow-1`, `flex-shrink-0`, `gap-*`

Position `position-relative/absolute/fixed/sticky`, `top-0`, `start-50`, `translate-middle`

Overflow `overflow-auto`, `overflow-hidden`, `overflow-scroll`

Visibility `visible`, `invisible`

Practice Exercises

1. Ek card banao jisme `p-4`, `shadow`, aur `rounded` classes ho.
2. `d-flex justify-content-between align-items-center` use kar ke ek navbar-jaisi row banao jisme left par logo aur right par menu items hon.
3. Ek notification badge banao `position-relative` parent aur `position-absolute top-0 start-100 translate-middle` child ke sath.

4. `d-none d-lg-block` aur `d-block d-lg-none` use kar ke mobile vs desktop content toggle karo.
 5. `vh-100` use kar ke ek full-screen hero section banao jisme `d-flex align-items-center justify-content-center` se content perfectly center ho.
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Active Recall — Bootstrap Utility Classes

┆ Khud se answer karne ki koshish karo, phir neech scroll karke check karo.

- Q1. Spacing classes ka naming pattern kya hai? `mt-5` aur `mx-auto` ka matlab kya hai?
 - Q2. `w-100` aur `vh-100` mein kya farq hai?
 - Q3. `rounded-circle` aur `rounded-pill` mein kya farq hai, aur `rounded-circle` use karne ke liye kya condition honi chahiye?
 - Q4. `shadow-sm`, `shadow`, aur `shadow-lg` mein farq kya hai?
 - Q5. `d-none` kya karta hai, aur `d-md-none` se ye kaise different hai?
 - Q6. Flexbox use karne ke liye pehla step kya hai (konsi class parent par lagani hoti hai)?
 - Q7. `justify-content-between` aur `justify-content-around` mein kya farq hai?
 - Q8. `flex-grow-1` aur `flex-shrink-0` kya karte hain?
 - Q9. `position-relative` aur `position-absolute` ka relationship kya hota hai jab ye sath use hote hain (parent-child)?
 - Q10. `overflow-hidden` aur `overflow-scroll` mein kya farq hai?
 - Q11. `d-none` aur `invisible` mein key difference kya hai?
 - Q12. Ek notification badge banane ke liye konsi classes combine hoti hain (`position-*`, `top-*`, `start-*`, `translate-*`)?
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Answers

<details> <summary>Answers dekhne ke liye click karein</summary>

A1. Pattern hai `{property}{sides}-{size}` — property `m` (margin) ya `p` (padding), sides `t/b/s/e/x/y` ya blank (all), size `0-5` ya `auto`. `mt-5` ka matlab margin-top bara, `mx-auto` ka matlab left-right margin auto (element ko horizontally center karta hai).

A2. `w-100` element ki width ko parent ke 100% ke barabar karta hai. `vh-100` element ki height ko **full viewport height** (100vh, पूरी screen height) ke barabar karta hai.

A3. `rounded-circle` element ko complete circle bana deta hai (isliye element ki width aur height barabar honi chahiye). `rounded-pill` element ko pill/capsule shape deta hai (bohat zyada border-radius, jaise buttons/badges).

A4. Teeno alag shadow sizes dete hain — `shadow-sm` chota/subtle shadow, `shadow` default/medium shadow, `shadow-lg` bara/prominent shadow.

A5. `d-none` element ko completely hide kar deta hai (`display: none`) — koi breakpoint condition nahi, hamesha hidden. `d-md-none` sirf medium screens ($\geq 768\text{px}$) aur uske upar element ko hide karta hai — chhote screens par element visible rehta hai.

A6. Parent element par `d-flex` class lagani hoti hai — usi se wo element ek flex container ban jata hai aur uske children flex items ban jate hain.

A7. `justify-content-between` items ke beech equal space deta hai lekin pehla item start edge par aur aakhri item end edge par touch karta hai (edges par space nahi). `justify-content-around` har item ke around equal space deta hai (edges par bhi thori space hoti hai, beech wali se kam).

A8. `flex-grow-1` item ko available extra space lene deta hai (remaining space fill karta hai). `flex-shrink-0` item ko shrink hone se rokta hai — apni original size maintain karta hai chahe container tight ho.

A9. `position-absolute` wala child apne nearest `position-relative` (ya kisi bhi positioned) parent ke relative position hota hai. Isliye parent par `position-relative`

lagana zaroori hai taake child usi ke andar accurately position ho, poori page ke relative nahi.

A10. `overflow-hidden` extra content ko simply chhupa deta hai (clip kar deta hai, koi scrollbar nahi). `overflow-scroll` hamesha scrollbar show karta hai, chahe content fit ho ya na ho.

A11. `d-none` element ko layout se completely hata deta hai — koi space occupy nahi karta. `invisible` element ko sirf visually hide karta hai, lekin uski space layout mein reserved rehti hai (jaisa element wahan ho hi, bas nazar nahi aa raha).

A12. `position-relative` parent par, aur child par `position-absolute top-0 start-100% translate-middle` (ya similar top/start values) — ye combination badge ko parent ke corner par exactly center-positioned rakhta hai.

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